Gradient Descent Intuition

repeat until convergence {

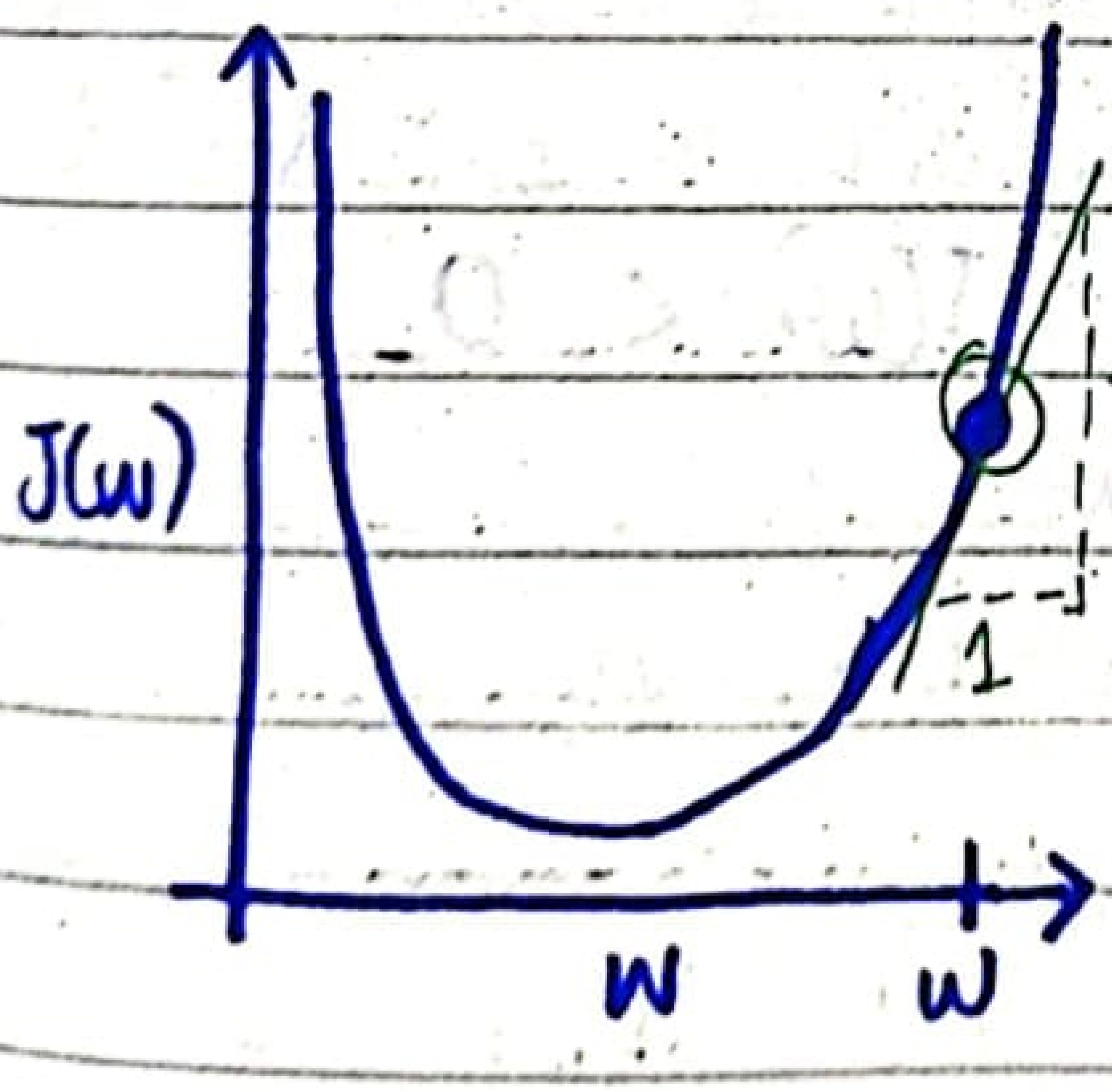
$$w = w - \alpha \boxed{\frac{d}{dw}(J(w,b))} \rightarrow \text{derivative}$$

learning rate

$$b = b - \alpha \frac{d}{db} J(w,b).$$

we'll use slightly simpler example: (minimizing)
with only one Parameter to see the algo:

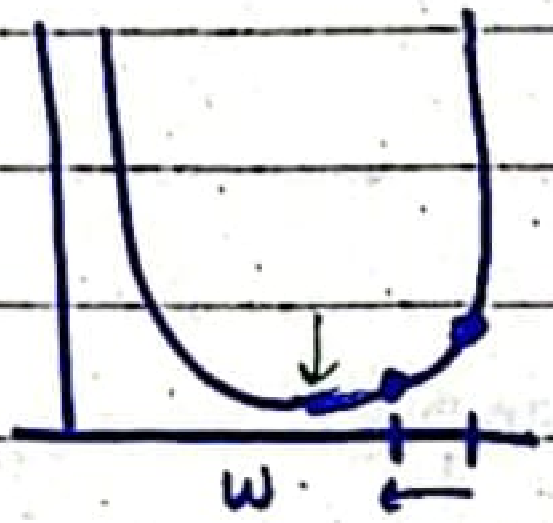
- $J(w)$,
- $w = w - \alpha \frac{d}{dw} J(w)$
- $\min_{w} J(w)$



A way to think a derivative at this point is to draw a tangent line.
The slope of this line is derivative of func J at this point.
To get slope you can make the triangle and divide height/width

& when tangent line is pointing to up & the right, slope is positive.

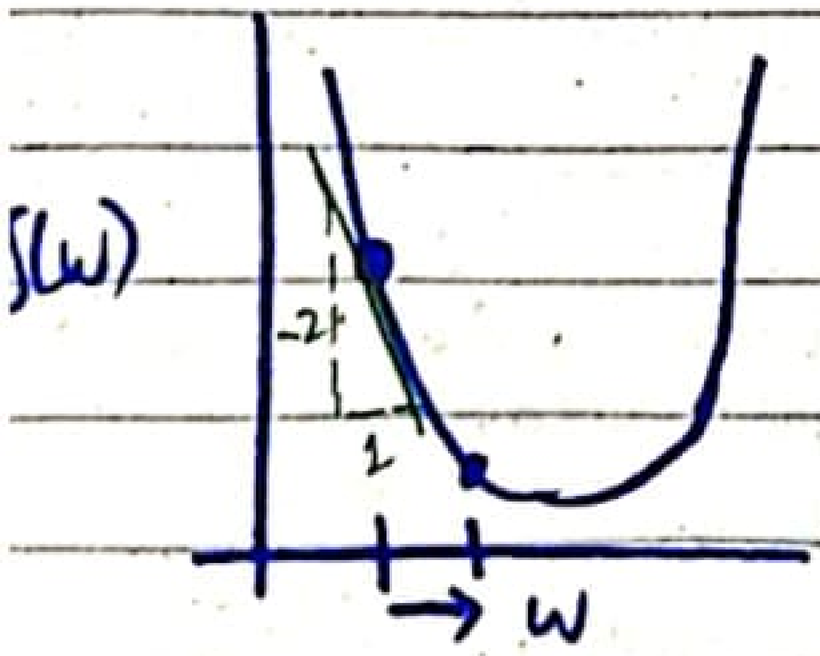
- So the derivative $\frac{d}{dw} J(w) > 0$.
- So $w := w - \alpha \cdot$ positive number.
  ↳ α always +ve
- now $w = w - \alpha \cdot p.n$
  ↳ w will update to smaller number.



So on graph, when we decrease w, it's right for our goal as J(w) also decrease as seen.

Another example:

diff starting point:



Now slope is negative as down & to the right. So derivative at this point $\frac{d}{dw} J(w) < 0$

- $w = w - \alpha \cdot$ negative no.

so w increases

as $-$ & $-$ gives $+$.